

2. (a) Chromium is one of the metals found in stainless steel. The equation shows how chromium is produced industrially by reacting chromium oxide with aluminium.



- (i) The reaction is highly exothermic.

Give the meaning of the term *exothermic*.

[1]

- (ii) During the reaction, oxidation and reduction happens.

I. Name the substance which is oxidised.

[1]

II. State what is meant by *reduction*.

[1]

- (iii) State what the equation tells you about the relative reactivities of chromium and aluminium.

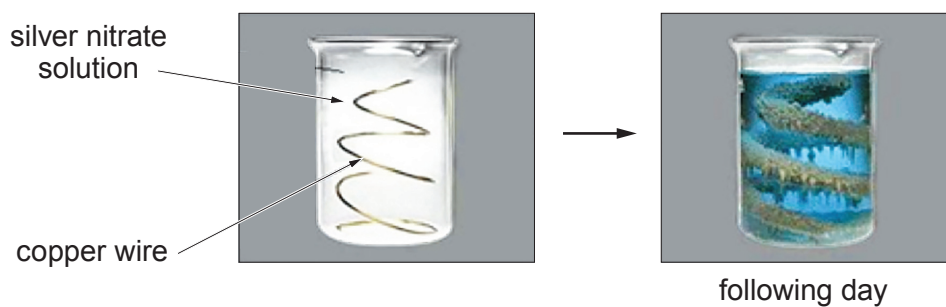
[1]



- (b) Copper is able to displace silver from a solution of silver nitrate. The equation for this reaction is given below.



A teacher demonstrated this reaction to her class. The photographs show the beaker before and after the reaction had taken place.



- (i) Explain how the changes show that this chemical reaction has taken place. [2]

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- (ii) Complete the symbol equation for the reaction by

- giving the formula for silver nitrate
- balancing the overall equation

[2]

